Learning from Data

- Can we learn about the world around us using data?
- Model building from data
  - Take data as input
  - Find patterns in the data
  - Summarize the pattern in a mathematically precise way
- Machine learning automates this model building.

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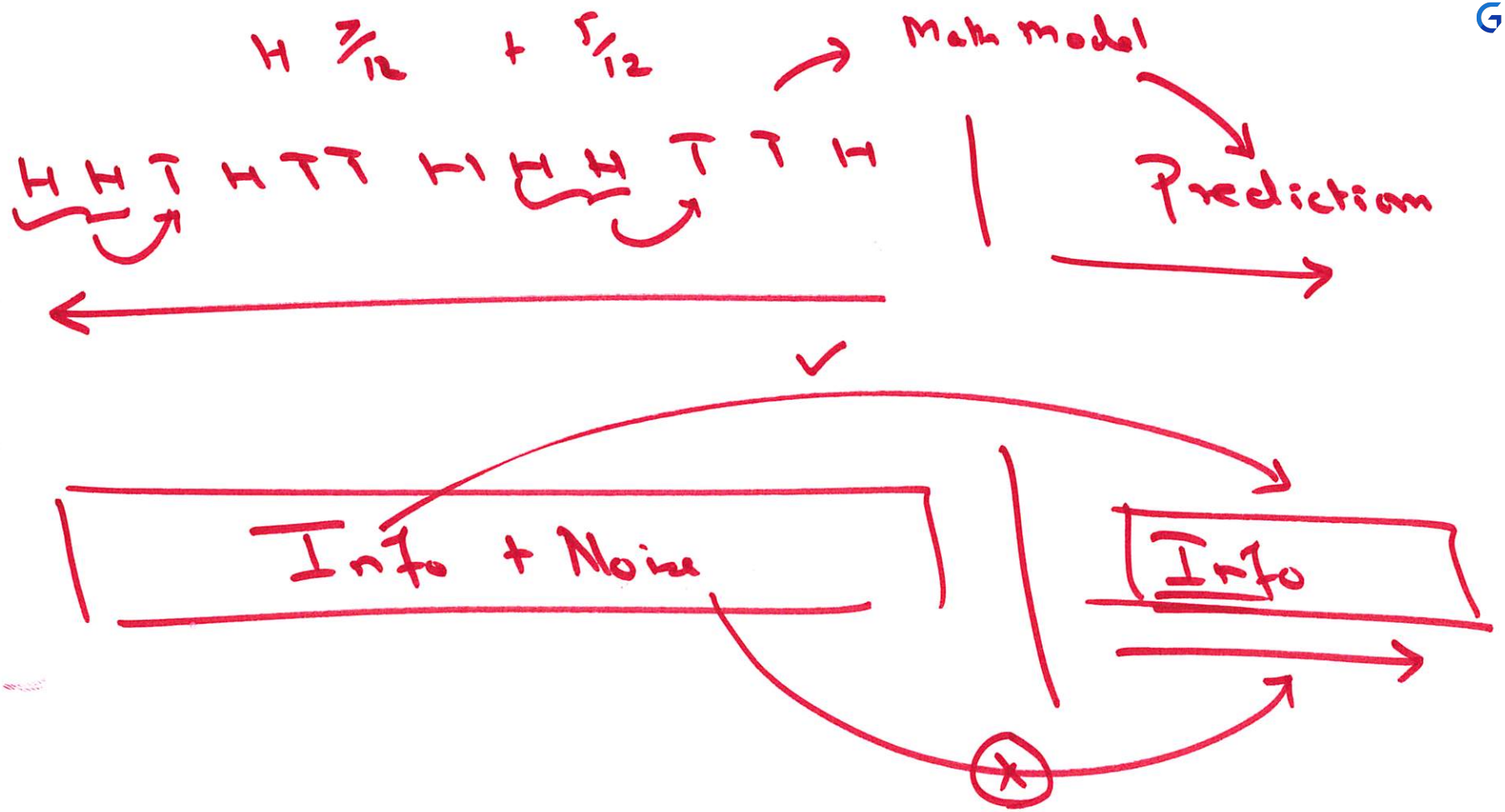

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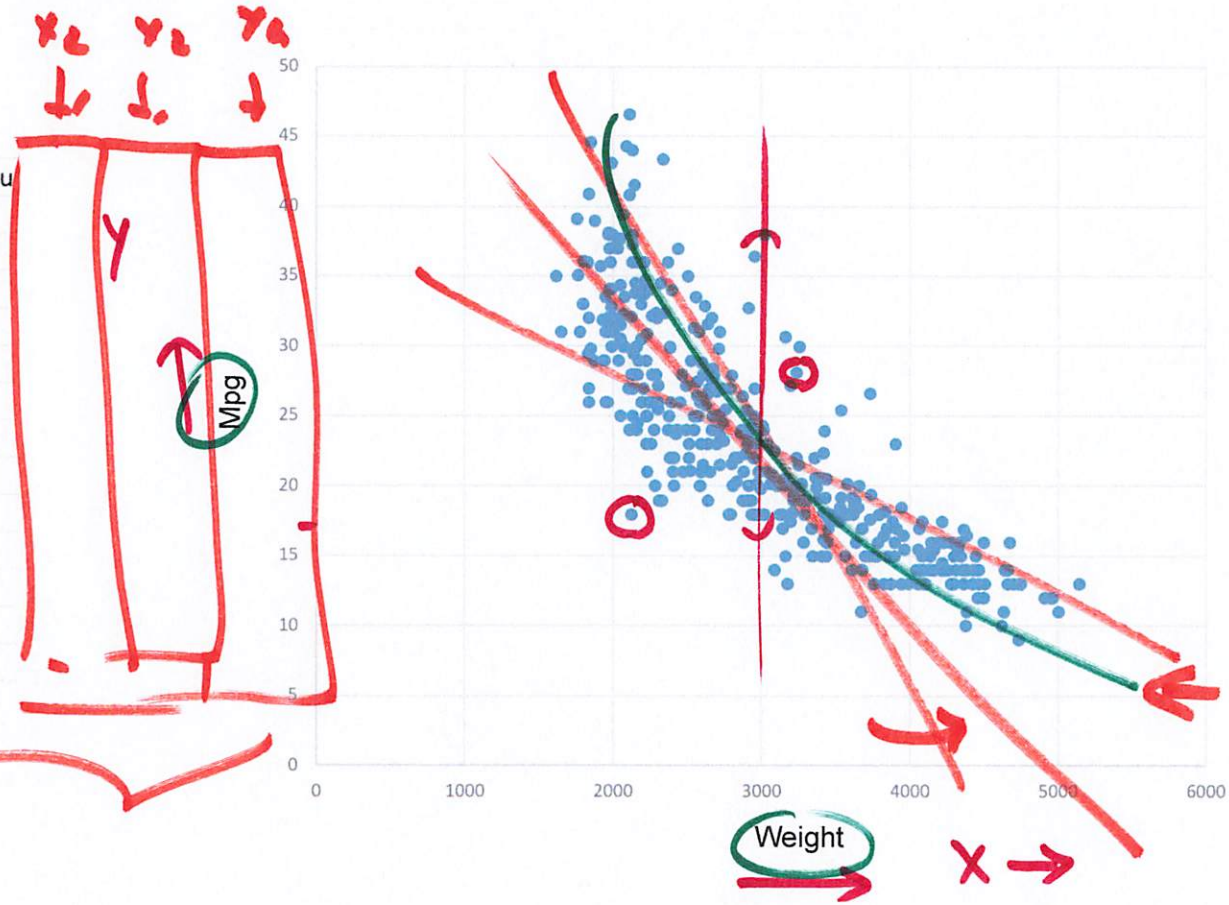
Linear Relations between two variables

learn from Data

- Do heavier cars have lower mileage?
- Can we use DATA to better understand relationships between the two variables: weight and mpg?

mpg	wt	car_name
18	3504	chevrolet chevelle malibu
15	3693	buick skylark 320
18	3436	plymouth satellite
16	3433	amc rebel sst
17	3449	ford torino
15	4341	ford galaxie 500
14	4354	chevrolet impala
14	4312	plymouth fury iii
14	4425	pontiac catalina
15	3850	amc ambassador dpl
15	3563	dodge challenger se
14	3609	plymouth 'cuda 340
15	3761	chevrolet monte carlo
14	3086	buick estate wagon (sw)
24	2372	toyota corona mark ii



Data Source: StatLib (http://lib.stat.cmu.edu/datasets/)

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⇒ Linear Regression

x	y	$\hat{y}$
⋮	⋮	⋮

need the best St. line

need the best a, b in

$$\boxed{y = a + bx}$$

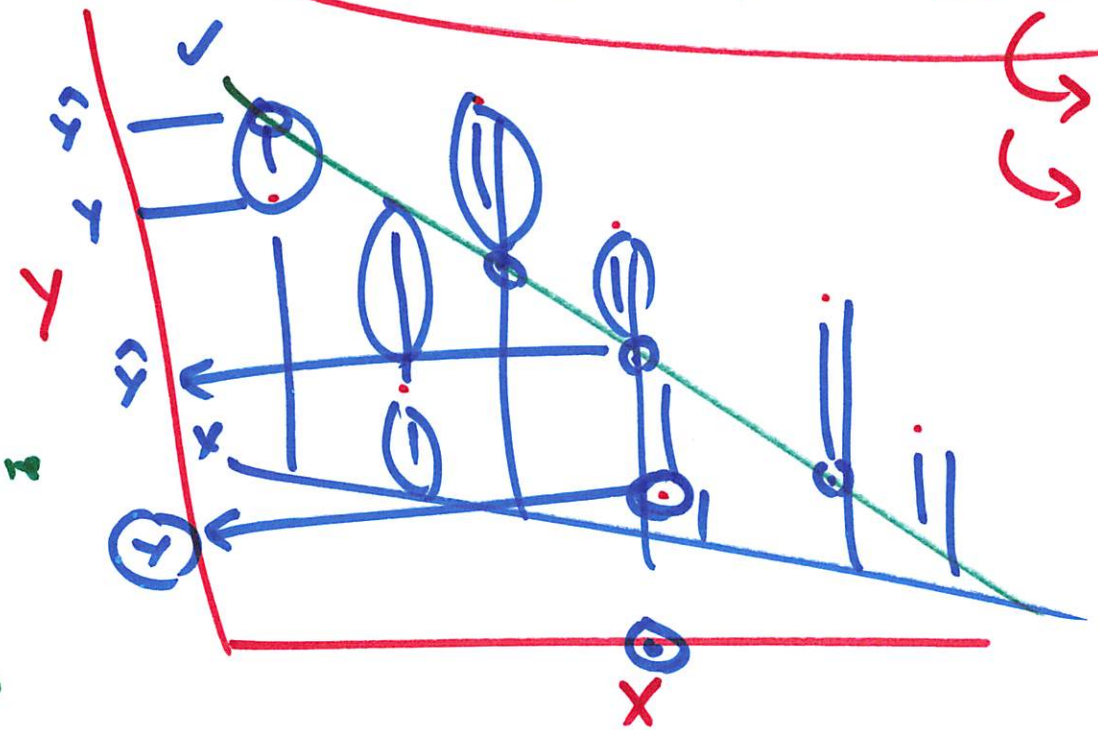
a → ? ✓

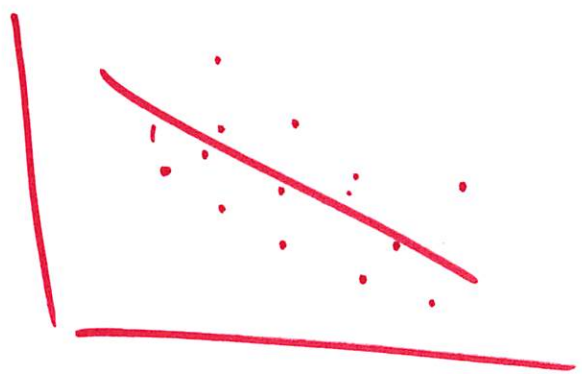
b → ? ✓

$$\text{Objective} = \sum [y - \hat{y}]^2$$

y → error

$\hat{y}$ → residual





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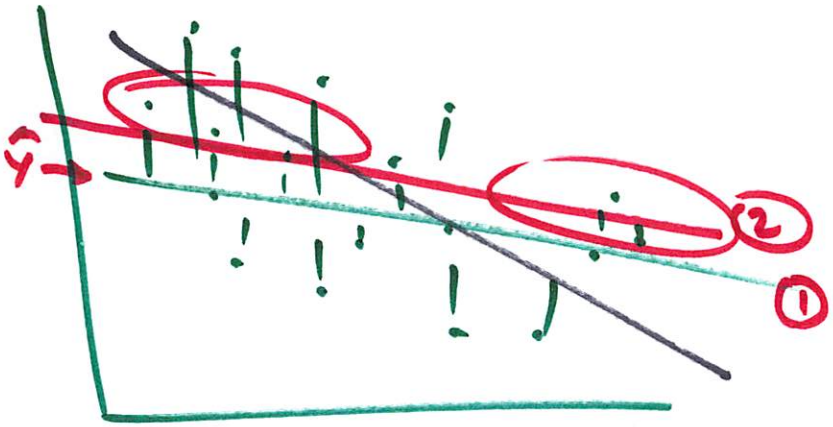

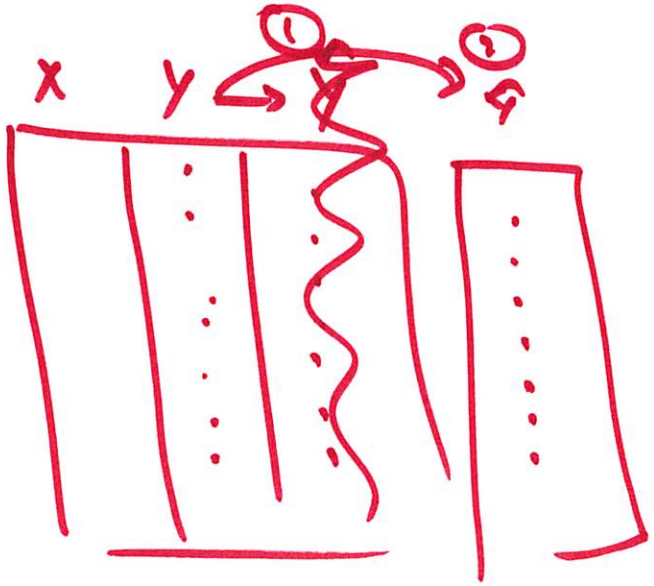

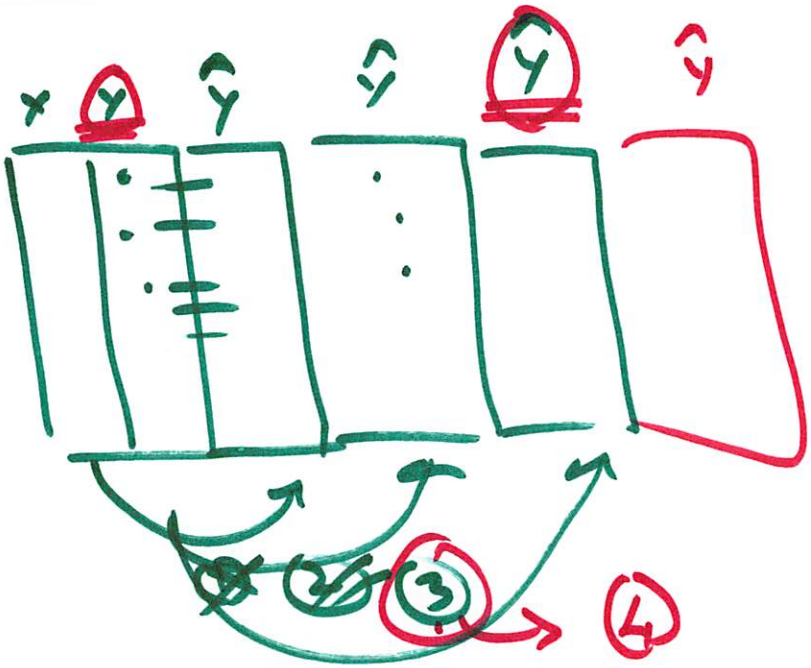

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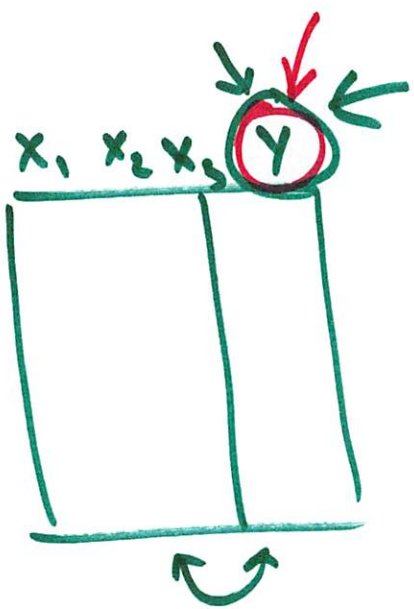
Machine learning

- Supervised learning
  - Regression - level
  - Classification - class
- Unsupervised learning
  - Clustering
  - Dim. Reduction
  - Market Basket Analysis.
- Reinforcement learning



$$\min_{a,b} \sum (Y - \hat{Y})^2$$

$$\boxed{\min_{a,b} \underbrace{\sum_i (Y_i - (a + b x_i))^2}}$$

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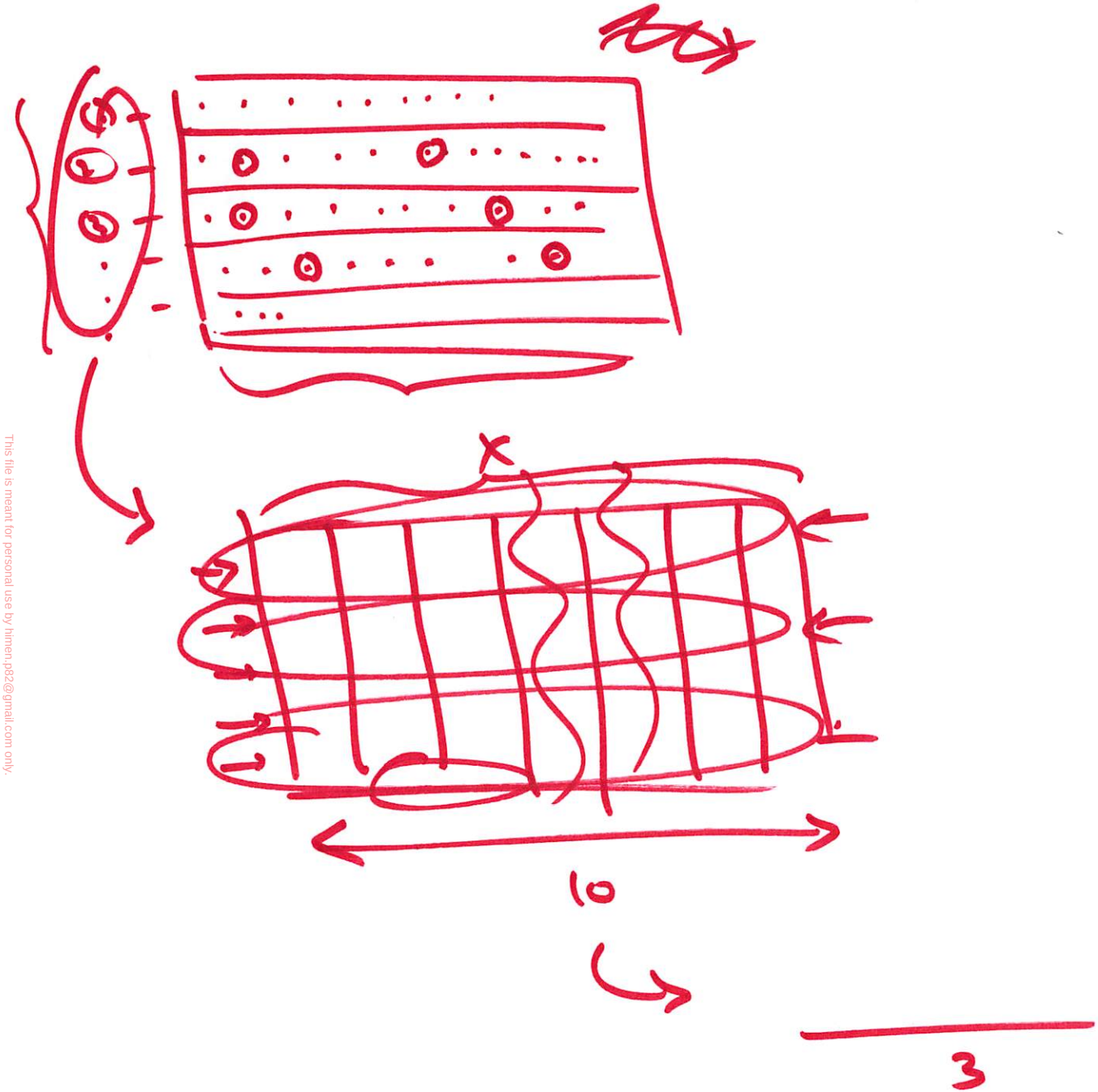

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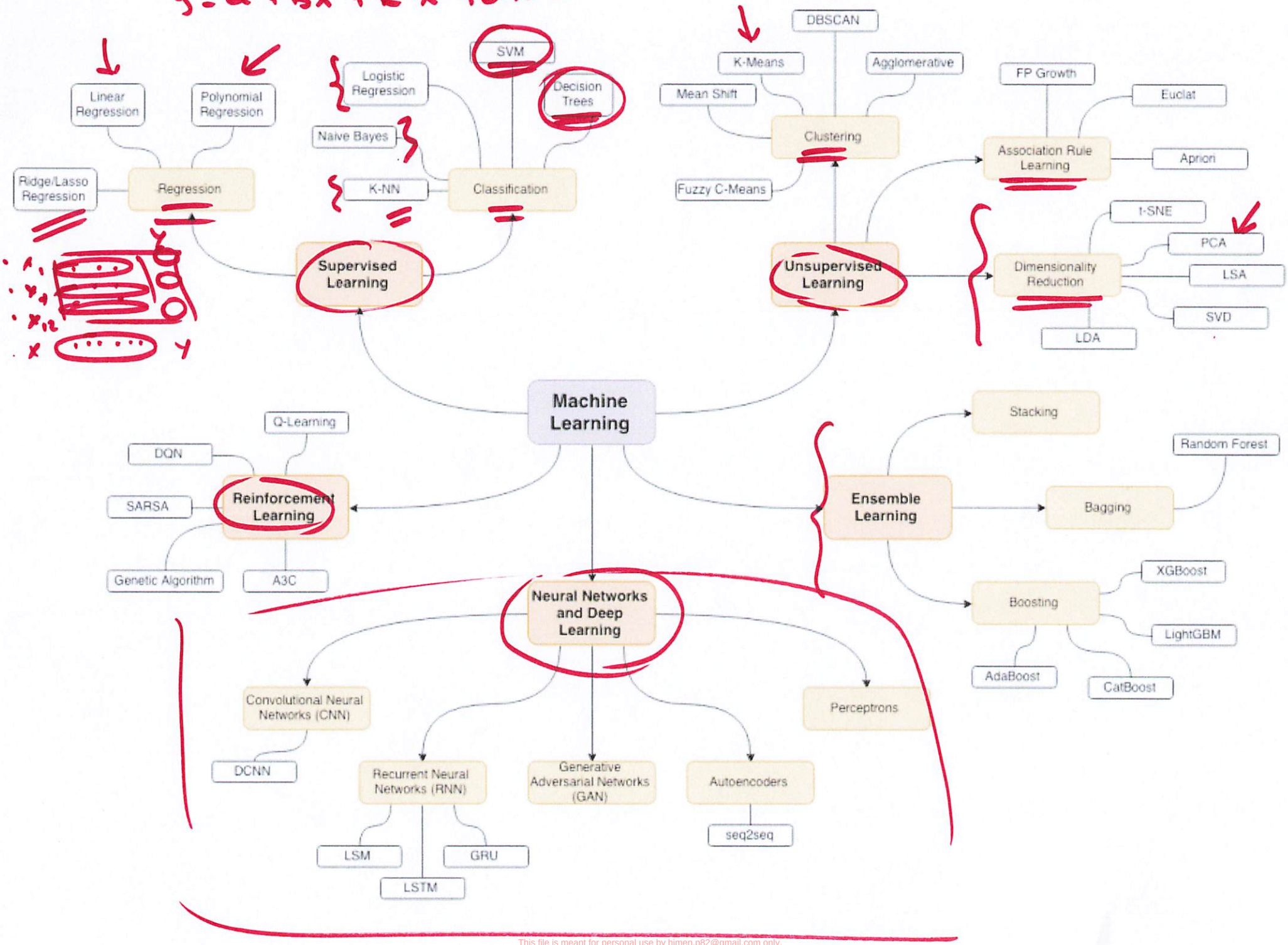
$y = a + bx + cx^2 + dx^3 \ldots$



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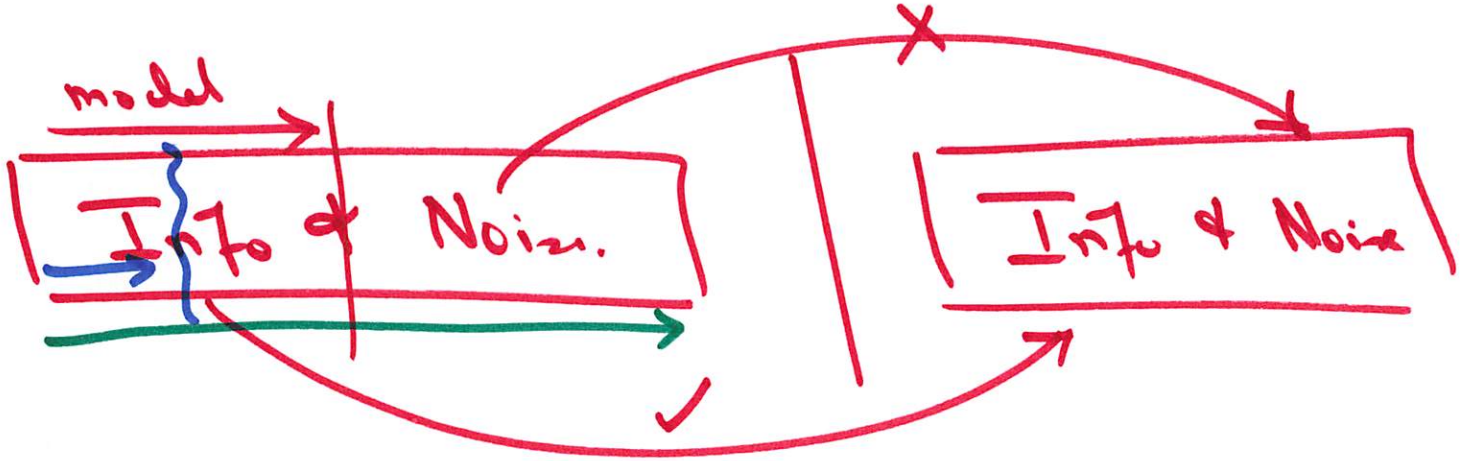
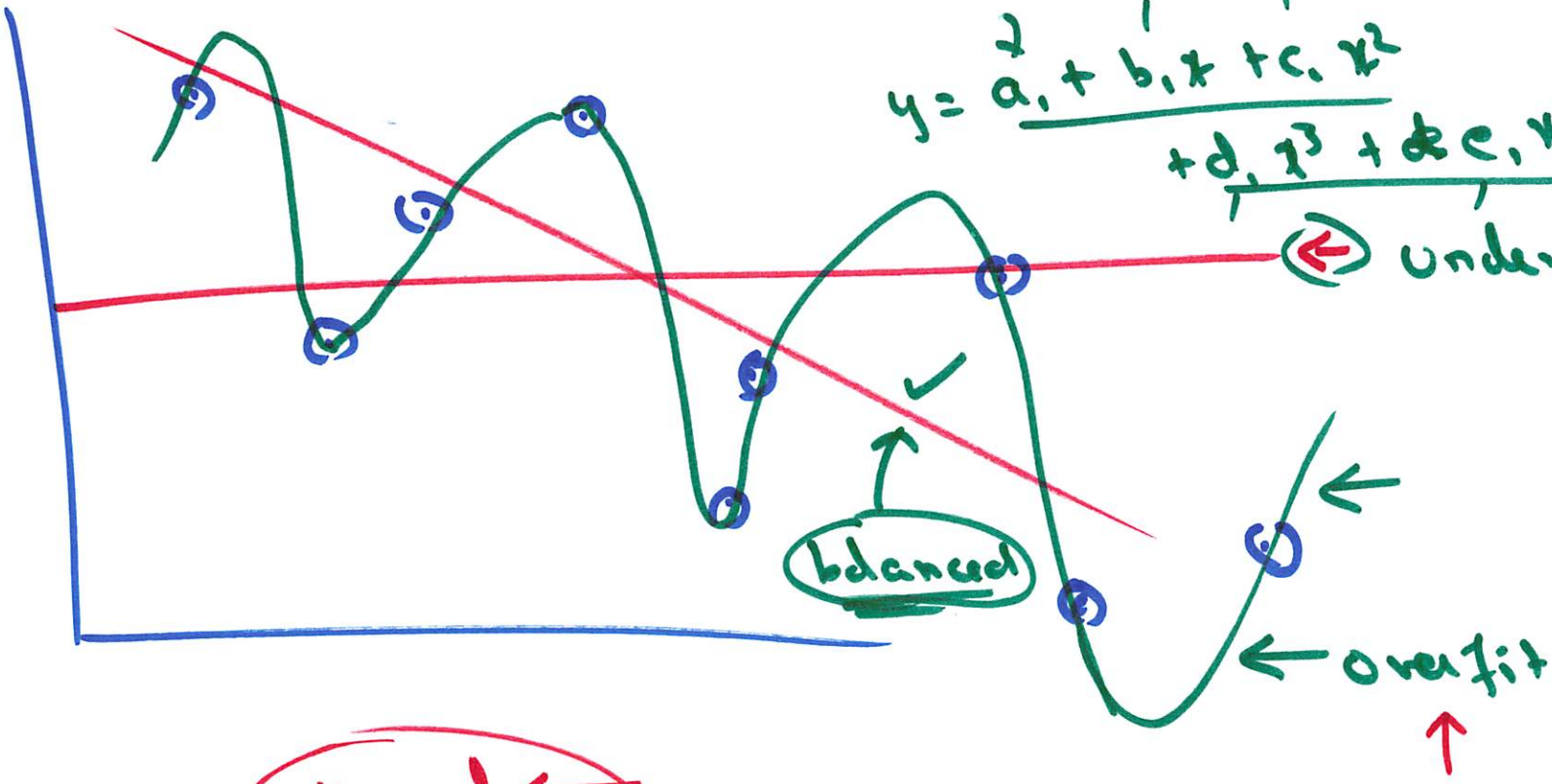


$y = a + bx$

$y = a_1 + b_1 x + c_1 x^2 + d_1 x^3 + e_1 x^4 + \dots$



$y = d$

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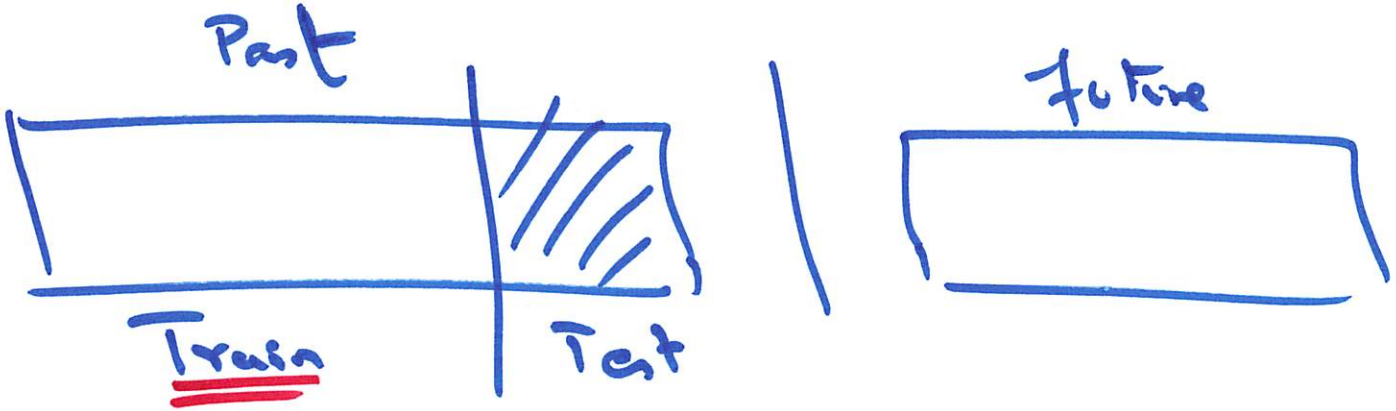

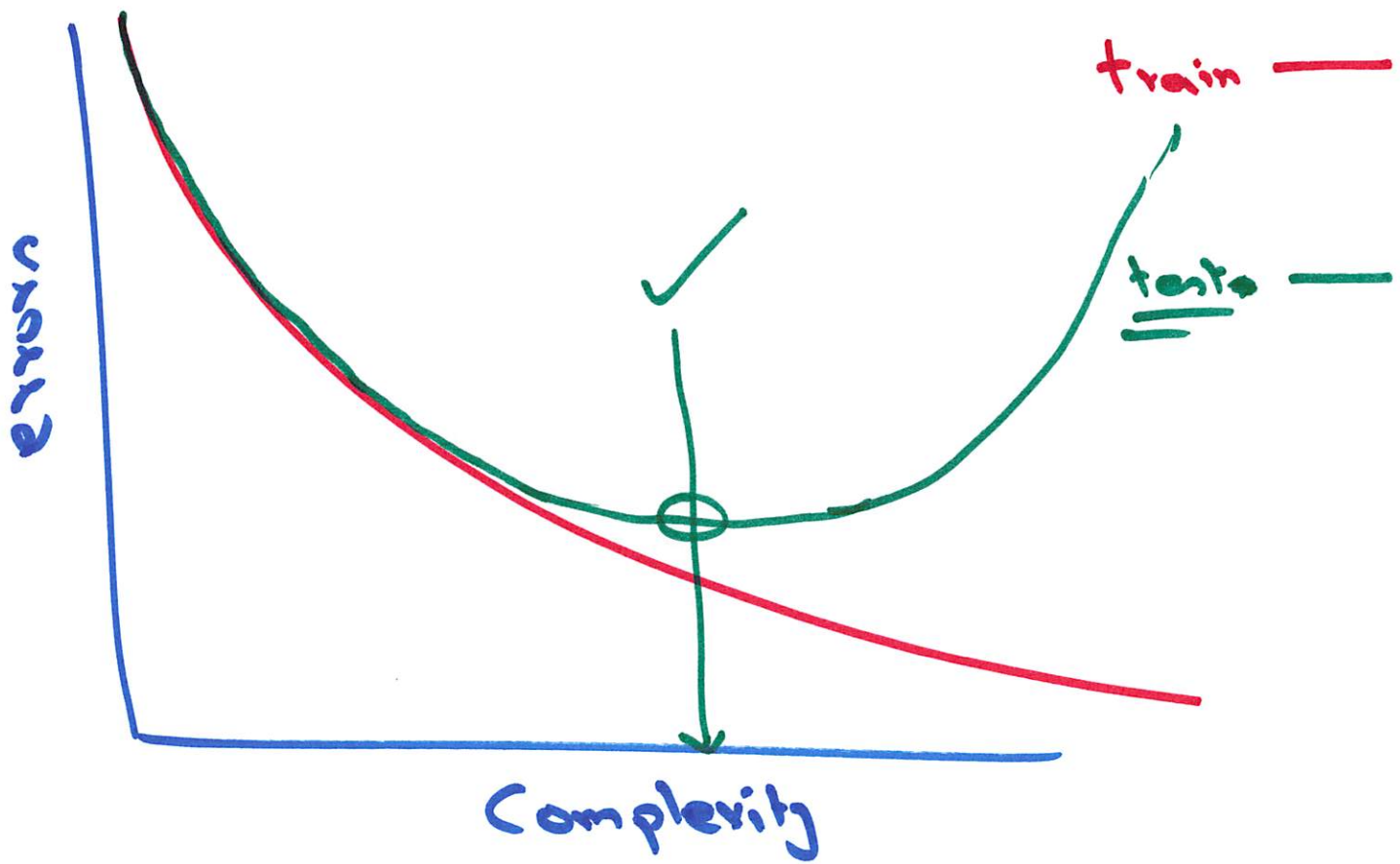

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